Abstract Locks and Transition DAGs: Repaired Consecutive-Firing Laws, the Dual-Ascent Barrier, and Ascent-Depth Bounds

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1 Abstract

Two abstract theories from the march-graph program are reported here, both [PROVED] in Lean. *Abstract locks* (Rounds 280/287) axiomatize the firing behavior of a lock frame $L = (W, c^\star, \tau, \lambda, \varepsilon)$: a window $W \subset \mathbb{Z}$ with traveling-hole, propagation-liveness, and edge-emptiness predicates, whose *fire set* is $\mathrm{FIRE}(L) = \{t \in W : \tau(t) \wedge \lambda(t) \wedge \varepsilon(t)\}$. The repaired consecutive-firing laws (Theorem 1, `consecT_gives_consec_fire`, `no_consecT_scattered`, `consec_window_no_consecT_not_full_no_consecT_gap_window`, `sharpness_witness`) hold in exactly the sharp form the original conjecture missed: the unconditional "no full lock without consecutive T's" clause was refuted by the explicit counterexample $W = \{0, 2\}$ (Q-24 OBJECTION), and we report the repair honestly. *Transition DAGs* (Rounds 279/288) study the Pareto-class transition graph $G = (C, E_d \cup E_p \cup \mathcal{L})$: the dual-ascent barrier (Theorem 2, `no_dual_target_demote`, `no_dual_source_promote`) is axiom-free pure order theory showing no demote edge targets a reflection dual of its source and no promote edge is sourced at a reflection dual of its target; the original stronger barrier D7 was demoted to [OPEN] by an explicit abstract non-entailment model (Q-26 OBJECTION). The L_s/C3 ascent-depth suite (Theorem 3, `dual_preserves_s_card`, `ascentDepth_le`, `ascentDepth_promote_mono`, `ascentDepth_eq_zero_iff`) bounds promotion chains in finite acyclic promote DAGs.

2 Introduction

The march-graph program tracks how Sidon 9-shapes evolve across spans $\sigma \to \sigma + 1$ through their $(|E|, |S|)$ Pareto classes. Two abstractions extracted from that concrete setting turned out to be worth proving in their own right. The first is the *abstract lock*: a lock frame strips away the Sidon content and keeps only the firing logic — a window, a shift, and three predicates deciding which times fire. The second is the *transition DAG*: a graph whose vertices are Pareto classes and whose edges record demotions, promotions, and persistence across span pairs, studied here through its reflection-dual structure and its promotion-chain depth.

Both theories passed through the program's rule-9 hostile review and came out changed: the abstract-lock theory lost its unconditional clause (killed by a two-element counterexample) and the transition-DAG theory lost its original barrier (killed by an abstract model). This paper's contribution is the *repaired* form of each theory, with the refutations kept on the record:

1. The sharp consecutive-firing laws for abstract locks (Theorem 1), with the Q-24 counterexample as a Lean sharpness witness.

2. The dual-ascent barrier in its axiom-free repaired form (Theorem 2), with the Q-26 model and the inapplicable conditional D7″ documented.

3. The L_s reflection-card preservation and C3 ascent-depth bounds (Theorem 3).

All are [PROVED]: bakkie full-library build exit 0, zero sorrys, axioms audited. The census audits that motivated them are [COMPUTED] evidence, never re-claimed as theorems.

3 Methods

3.1 Abstract lock definitions

We use the R280 definitions verbatim (supplied datum). A *lock frame* is a tuple $L = (W, c^\star, \tau, \lambda, \varepsilon)$ where $W \subseteq \mathbb{Z}$ is finite and nonempty (the *firing window*), $c^\star \in \mathbb{Z}$ is the *lock shift*, $\tau : W \to \{T, F\}$ is the *traveling-hole predicate* ($\tau(t)$: the traveling hole is present at time t), $\lambda : W \to \{T, F\}$ is the *propagation-liveness predicate* ($\lambda(t)$: propagation is live at time t), and $\varepsilon : W \to \{T, F\}$ is the *edge-emptiness predicate* ($\varepsilon(t)$: $\mathrm{edge}^Y_{t-c^\star} = \varnothing$ — the data). The *fire set* is

$$\mathrm{FIRE}(L) = \{t \in W : \tau(t) \wedge \lambda(t) \wedge \varepsilon(t)\}. \tag{1}$$

A frame is *complete* when $\tau \equiv T$ and $\lambda \equiv T$ on W. A complete frame is *full* if $\mathrm{FIRE}(L) = W$, *partial* if $\varnothing \subset \mathrm{FIRE}(L) \subset W$, and *void* if $\mathrm{FIRE}(L) = \varnothing$; statuses are exhaustive and pairwise disjoint by construction. An abstract *lock* is a full or partial frame ($\mathrm{FIRE} \neq \varnothing$).

We write *consec*T for "ε has consecutive T's": $\exists n \in \mathbb{Z},\ \varepsilon(n) = \varepsilon(n+1) = T$. *Consecutive firing* means FIRE contains two consecutive integers; *scattered firing* means it contains no two consecutive integers; a *gap window* is a W containing no two consecutive integers. In the Lean module (complete frames), $\texttt{fire}(W, \varepsilon) = \{t \in W : \varepsilon(t)\}$.

3.2 Transition DAG definitions

The *transition DAG* (R279, supplied datum) is $G = (C, E_d \cup E_p \cup \mathcal{L})$, where C is the 195 Pareto classes (points of $\mathbb{N}^2$): E_d is the 24 demote edges, directed demoted→demoter ("yield") and each labeled by its σ-pair; E_p is the 16 promote edges, directed blocker→promoted (13 promote events, 3 with 2 blockers); $\mathcal{L}$ is the 14 persistence loops (c, c). The *persistence window* of c is a maximal contiguous interval $I \subseteq W(c)$, where $W(c) = \{\sigma \in [80, 183] : c \in \Pi(\sigma)\}$. A *merge* (per pair) is a demoter q with indegree ≥ 2 in that pair's demote edges; a *split* (per pair) is a demoted c with outdegree ≥ 2. A *dominance flip-flop* is a 2-cycle $c \rightleftarrows q$ in $E_d \cup E_p$; the census has exactly one, $(14, 4) \rightleftarrows (14, 5)$ (supplied datum).

Dual ascent (R288): d is a reflection dual of class c at span σ iff $d = \mathrm{class}(\bar{A})$ for some span-σ witness shape A of c ($\bar{A} = \{\sigma - a\}$, T2′/Lemma 5, cited as given). In Lean, with $\texttt{Class} := \mathbb{N} \times \mathbb{N}$:

$$\texttt{IsDual}(d, c) :\Leftrightarrow d.2 = c.2 \ \wedge\ c.1 \leq d.1, \tag{2}$$

duals sit e-above-or-equal; $\texttt{IsDemoteEdge}(c, q) :\Leftrightarrow q.1 \leq c.1 \wedge q.2 \leq c.2 \wedge q \neq c$; $\texttt{IsPromoteEdge}(q, a) :\Leftrightarrow q.1 \leq a.1 \wedge q.2 \leq a.2 \wedge q \neq a$.

For the ascent-depth suite (R292, folded R298): over a finite vertex type V with promote relation E_p, $\texttt{IsPromoChain}(c, n)$ is $\exists f : \mathrm{Fin}(n{+}1) \to V,\ f(0) = c$ with consecutive E_p edges; $\texttt{PromoAcyclic}$ is $\forall c,\ \neg\,\mathrm{TransGen}\, E_p\, c\, c$ (R292's D1b); and

$$\texttt{ascentDepth}(c) = \max\{n : \texttt{IsPromoChain}(c, n)\}, \tag{3}$$

taken over $0 \leq n \leq |V| - 1$ (nonempty: the trivial chain). The L_s mechanism is $W(\bar{A}) = \sigma - W(A)$ as sets, $w \mapsto \sigma - w$ bijecting $[0, \sigma]$; its order-theoretic core is reflection card preservation.

Table 1: Formalized theorems, all [PROVED]. Module paths are repo-relative. The two barrier theorems are axiom-free.

Lean theorem	content	axioms
`Mathproofs/AbstractLock.lean`		
`consecT_gives_consec_fire`	R287 (a), forward	standard
`no_consecT_scattered`	R287 (a), contrapositive	standard
`consec_window_no_consecT_not_full`	R287 (c)(i), sharp	standard
`full_no_consecT_gap_window`	R287 (c)(ii), equivalent	standard
`sharpness_witness`	premise necessity, $W = \{0, 2\}$	standard
`Mathproofs/TransDAG.lean`		
`no_dual_target_demote`	D7′(a), dual-ascent barrier	none
`no_dual_source_promote`	D7′(b), dual-ascent barrier	none
`dual_preserves_s_card`	L_s reflection card	standard
`ascentDepth_le`	C3(i) depth bound	standard
`ascentDepth_promote_mono`	C3(ii) monotonicity	standard
`ascentDepth_eq_zero_iff`	C3(iii) zero characterization	standard

3.3 Formalization note

The machine-checked content lives in the repository `derekblair/math-proofs`, Lean 4.34.1 with Mathlib pinned at commit `d13f23b`. Verification config: full-library `lake build` on the bakkie fleet machine (2026-10-11, 11:12 PDT): "Build completed successfully (17936 jobs)" — exit 0, 0 errors, 0 `sorry`s, 0 `admit`s in the log. The R310 label move took both modules from [PROVED*] to [PROVED] (charter rule 4). Modules are md5-identical local ↔ bakkie (`AbstractLock.lean` `54de580f...`; `TransDAG.lean` `360a3271...`), each registered exactly once per side; zero sorrys in both (grep-verified). Axiom audits: the two dual-ascent barrier theorems are *axiom-free* (0 axioms); all other named theorems: `[propext, Classical.choice, Quot.sound]` (standard Mathlib set, no `sorryAx`). Each module's header points at this paper's slug `papers/abstract-lock-transdag-October-11-2026/`; this paper points back at the modules and the exact theorem names of Table 1.

The hostile-review scope is part of the claim. Q-32 (Bunny, quoted verbatim 2026-10-11 01:29 PDT): "Q-32: (a) SURVIVES, (b) SURVIVES, (c) SURVIVES for complete finite nonempty frames with integer adjacency. Clause (c) sharpness is a window-only guarantee uniformly over ε." Q-33 (Bunny, quoted verbatim): "Q33 (a): D7′ SURVIVES for full-spectrum Pareto endpoints, or explicit all-duals ascent. OBJECTION to applying it to sampled-census Pareto endpoints from Q22 alone: the forbidden dual may be absent from that census." and "Q-33: OBJECTION to census application. D7′/D7″ SURVIVE with full-spectrum Pareto endpoints or proved all-duals ascent. Complementary abstract models survive." The 195-class Sidon census application is therefore NOT claimed. Q-36 (Bunny, quoted verbatim 2026-10-11 03:14–03:35 PDT): "L_s SURVIVES" and "C3 SURVIVES (finite promotion DAGs, bound $|C| - 1$)", scoped to finite promotion DAGs; C1/C2/C4 received OBJECTION and are not formalized here.

4 Results

Theorem 1 (Repaired consecutive-firing laws; PROVED)**.** *For a complete finite nonempty frame with integer adjacency:*

(a) (`consecT_gives_consec_fire`) If $n, n+1 \in W$ and $\varepsilon(n) = \varepsilon(n+1) = T$, then FIRE *contains two consecutive integers — a consecutive-firing lock. Contrapositive (`no_consecT_scattered`):*

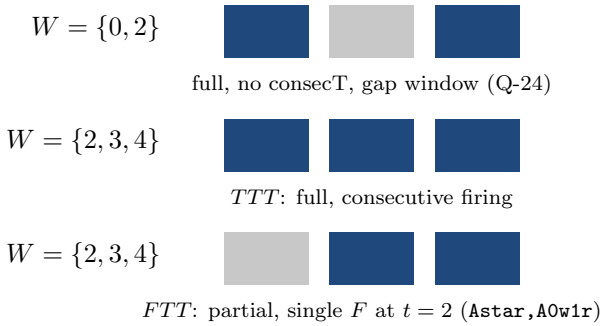


Figure 1: Firing patterns on concrete frames (supplied datum, R280 census): the Q-24 witness $W = \{0, 2\}$ (top) is full yet has no consecutive firing times; $W = \{2, 3, 4\}$ with $\tau \equiv \lambda \equiv T$ realizes full/TTT (Astar,E3) and partial/FTT (Astar,A0w1r, Astar,N5w1). Blue cells fire; gray cells do not.

if ε has no consecutive T's, firing is at most scattered, never a consecutive-firing lock.

$$\big(n, n{+}1 \in W \ \wedge \ \varepsilon(n) = \varepsilon(n{+}1) = T\big) \implies \exists m : \ m, m{+}1 \in \mathrm{FIRE}. \tag{4}$$

(b) (Failure admitted.) The original clause — "no consecutive T's in ε implies never a full lock on $|W| \geq 2$" — is FALSE as stated. Q-24's OBJECTION stands; see Proposition 1.

(c)(i) (`consec_window_no_consecT_not_full`, sharp) If W contains two consecutive integers and ε has no consecutive T's, the frame is NOT full (and firing is scattered). The consecutive-integer premise is NECESSARY.

$$\big(n, n{+}1 \in W \ \wedge \ \neg\exists k : \varepsilon(k) = \varepsilon(k{+}1) = T\big) \implies \neg\mathrm{full}. \tag{5}$$

(c)(ii) (`full_no_consecT_gap_window`, equivalent form) If the frame is full and ε has no consecutive T's, then W contains no two consecutive integers — a gap window.

$$\big(\mathrm{full} \ \wedge \ \neg\exists k : \varepsilon(k) = \varepsilon(k{+}1) = T\big) \implies \neg\exists n : \ n, n{+}1 \in W. \tag{6}$$

Full $\iff \varepsilon \equiv T$ on W (R280 Theorem 2(i), untouched) was over-extended by the original T3(iii): $|W| \geq 2$ yields two *distinct* points of W, not two *consecutive* integers — the quantifier slip located by the R287 repair (supplied datum).

Proposition 1 (Q-24 counterexample; PROVED as sharpness witness)**.** *The frame $W = \{0, 2\}$ with $\tau \equiv \lambda \equiv \varepsilon \equiv T$ is complete and full (*FIRE $= \{0, 2\} = W$*), has no consecutive firing times, and ε has no consecutive T's. The original "no-full-on-$|W| \geq 2$-without-consecutive-T's" clause is false. The Lean `sharpness_witness` re-derives this: $W = \{0, 2\}$, ε true exactly on W — FULL, no consecutive T's anywhere, W with no consecutive integers — proving the consecutive-integer premise of (c)(i) cannot be dropped.*

The Q-24 OBJECTION (Bunny, quoted verbatim 2026-10-10 22:00 PDT): "T3(iii) counterexample $W = \{0, 2\}$, $\tau = \lambda = \varepsilon = \mathit{true}$ — complete+full, no consecutive firing times." The review's mechanism: full $\iff \varepsilon \equiv T$ on W is true, but the step "$\varepsilon \equiv T$ on W with $|W| \geq 2$ implies ε has consecutive T's" needs W to contain consecutive integers, which $|W| \geq 2$ does not supply.

The repaired theory was audited computationally (R287, [COMPUTED]×2, zero shared code): the 472-frame universe — all complete frames with $W \subseteq \{0, \ldots, 5\}$, $1 \leq |W| \leq 4$, every ε-pattern — contains 134 consecutive-firing frames, 20 gap-full frames (full with no consecutive T's, including the reviewer's $W = \{0, 2\}$), and 36 consecutive-window-full frames; path 1 ran

Table 2: The 472-frame R287 census (supplied [COMPUTED]×2 datum).

class	count
total complete frames ($W \subseteq \{0,\ldots,5\}$, $1 \le \lvert W\rvert \le 4$, all ε)	472
consecutive-firing frames	134
gap-full frames (full, ε with no consecT)	20
consecutive-window-full frames	36
verification path 1 checks	1,398 (all pass)
verification path 2 checks	1,262 (all pass)

1,398 checks ALL PASS, path 2 ran 1,262 checks ALL PASS (supplied datum). The R280 concrete instances (supplied datum) are: $(\texttt{Astar}, \texttt{E3}) \to (T,F,C) = (0,0,0)$, full, pattern TTT; $(\texttt{Astar}, \texttt{AOw1r})$, $(\texttt{Astar}, \texttt{N5w1}) \to (1,0,1)$, partial, pattern FTT with the single F at $t=2$ ($\text{edge}_{-9} = \{184\}$, $\text{edge}_{-1} = \{184\}$). The void pattern FFF is absent from the census — an abstract prediction of the theory, not an input.

Theorem 2 (D7′ dual-ascent barrier; PROVED, axiom-free)**.** *Let IsDual, IsDemoteEdge, IsPromoteEdge be as in (2). Then:*

(a) (`no_dual_target_demote`) No demote edge targets a reflection dual of its source:

$$\texttt{IsDemoteEdge}(c,q) \ \wedge\ \texttt{IsDual}(q,c) \implies \bot. \tag{7}$$

(b) (`no_dual_source_promote`) No promote edge is sourced at a reflection dual of its target:

$$\texttt{IsPromoteEdge}(q,a) \ \wedge\ \texttt{IsDual}(q,a) \implies \bot. \tag{8}$$

The proof is pure finite order theory (T7′+A2): for (a), suppose $q = (e', s)$ is a σ-dual of $c = (e, s)$; A2 gives the same s, T7′ ($c \in \Pi(\sigma)$) gives $e' \ge e$; the demote orientation (D1, 24/24 scripted) gives $q \le c$, $q \ne c$, hence $e' \le e$; so $e' = e$, $q = c$, contradicting $q \ne c$. Part (b) is symmetric. Both theorems are *axiom-free* (0 axioms, R294). The R288 census audit ([COMPUTED]×2, zero shared code) found 0 violations of D7′(a) and 0 of D7′(b) — 0 dual-related edges anywhere over 294 Pareto instances / 24 demote + 16 promote edges (supplied datum).

Proposition 2 (Original D7 demoted to [OPEN)**.** *] The original D7 ("no DAG edge relates a class to its reflection dual as distinct Pareto nodes", a T8′ corollary conditional on T2′) does not follow from fixed-span T7′/T8′. Q-26's abstract non-entailment model (reconstructed in `r288_reviewer_model.py`, all asserts pass; supplied datum): $S_\sigma = \{(5,5),(6,5)\}$ with swapped duals, $S_{\sigma+1} = \{(6,5)\}$ self-dual. Verified: T7′ holds at both spans (Pareto duals have $e' \ge e$); T8′ holds at both spans (no distinct co-Pareto duals); T5 yields the promote edge $(5,5) \to (6,5)$ with blocker $(5,5)$; $(6,5)$ IS the σ-dual of $(5,5)$ — so a DAG edge relates a class to its reflection dual as distinct Pareto nodes, and D7's conclusion FAILS. The broken step: "hence never Pareto" drops T8′'s same-σ qualifier — Pareto status is span-relative; fixed-span premises cannot constrain cross-span Pareto status. A second model realizes the demote-side half: $S_\sigma = \{(6,5),(7,5)\}$, $S_{\sigma+1} = \{(5,5),(6,5)\}$, demote edge $(6,5) \to (5,5)$ with $(6,5)$ a $(\sigma{+}1)$-dual of $(5,5)$.*

The conditional D7″ — dual non-domination H: "no Pareto $q \in \Pi(\sigma)$ has a DISTINCT dual $d \in S(\sigma)$ with $q \le d$" implies the full D7 — is recorded as [PROVED* conditional] but *inapplicable*: H is FALSE on the census (39 violations, e.g. $\sigma = 81$: $(37,1)$ Pareto dominates its dual $(47,1) \in S(81)$; H_{sym} fails with 3 violations; 15 self-dual Pareto instances; supplied datum). The complementary halves (promote target = dual of source; demote source = dual of target) stay [OPEN] — abstract models realize both, so the full D7 is contingent, not a theorem.

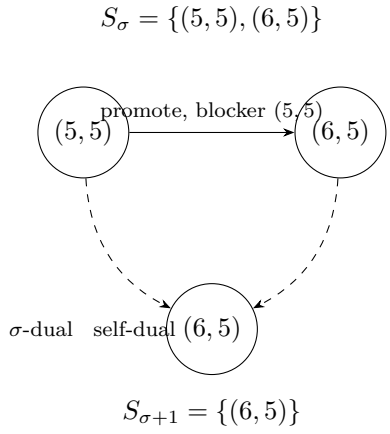


Figure 2: Q-26's abstract non-entailment model (Proposition 2): T7′ and T8′ hold at both spans, yet the promote edge $(5,5) \to (6,5)$ relates a class to its reflection dual as a distinct Pareto node. Fixed-span premises cannot constrain cross-span Pareto status, so the original D7 is [OPEN], not a theorem (supplied datum, R288 dossier).

Theorem 3 (L_s/C3 ascent-depth bounds; PROVED)**.** *Over a finite vertex type V with promote relation E_p and* `PromoAcyclic` *(R292's D1b):*

*(L_s) (**`dual_preserves_s_card`**) Reflecting a finset in a bound preserves card: for all σ and $W \subseteq \mathbb{N}$ finite with $w \le \sigma$ on W,*

$$\big(W.\mathrm{image}(w \mapsto \sigma - w)\big).\mathrm{card} = W.\mathrm{card}, \tag{9}$$

the order-theoretic core of "duals preserve s" ($W(\bar{A}) = \sigma - W(A)$, $w \mapsto \sigma - w$ bijecting $[0,\sigma]$).

*(C3-i) (**`ascentDepth_le`**)* $\texttt{ascentDepth}(c) \le |V| - 1$ *for all c.*

*(C3-ii) (**`ascentDepth_promote_mono`**) $E_p(q,a)$ implies* $\texttt{ascentDepth}(a) + 1 \le \texttt{ascentDepth}(q)$*: ascent depth strictly increases against the promote direction (prepend q to a longest chain from a; q is fresh, else a positive closed walk against acyclicity).*

*(C3-iii) (**`ascentDepth_eq_zero_iff`**)* $\texttt{ascentDepth}(c) = 0 \iff \forall a,\ \neg E_p(c,a)$*: zero depth iff c sources no promote edge.*

Theorem 3 is R292's L_s/C3 suite folded into `TransDAG.lean` in R298, [PROVED] in R310 with standard axioms. Q-36 (Bunny, quoted verbatim 2026-10-11 03:14–03:35 PDT): "L_s SURVIVES" and "C3 SURVIVES (finite promotion DAGs, bound $|C|-1$)", scoped to finite promotion DAGs; C1/C2/C4 received OBJECTION and are not claimed.

5 Discussion

Significance. The abstract-lock theory shows how little is needed to force firing structure: from a window and three predicates, the repaired laws (Theorem 1) recover the exact boundary between consecutive-firing, scattered-firing, and gap-window regimes, and the Q-24 witness proves the boundary is sharp. The 472- frame census (Table 2) is a genuine discovery of the abstract theory: 20 gap-full frames exist, and the void FFF pattern never occurs. The dual-ascent barrier (Theorem 2) is the strongest statement that survives the Q-26 objection — axiom-free pure order theory, audited at 0 violations over 294 Pareto instances. The L_s/C3 suite turns the transition DAG's promote edges into a well-founded ascent-depth measure bounded by $|C|-1$.

The refutation record. We keep both objections visible because the repairs are the paper's real content. The original T3(iii) fell to a two-point window ($W = \{0, 2\}$); the surviving content is the sharp window-conditional form (c)(i)/(c)(ii), and the Lean `sharpness_witness` is the counterexample promoted into a theorem about premise necessity. The original D7 fell to an abstract model with swapped duals; the surviving content is the axiom-free barrier D7′ (the two halves that follow from T7′+A2 alone), while the conditional D7″ is recorded but inapplicable (H false on the census) and the complementary halves stay [OPEN]. This is the honest boundary: D7′ is not a recovery of D7.

Limitations. (i) The dual-ascent barrier's premises (T7′ with s-preservation at both spans) are abstract hypotheses; the 195-class Sidon census application is explicitly out of scope per Q-33. (ii) The abstract-lock census covers only $W \subseteq \{0, \ldots, 5\}$ with $|W| \leq 4$ — the sharpness claims generalize by proof, but the empirical pattern (void FFF absent) is a census observation, not a theorem. (iii) Q-32's scope is "complete finite nonempty frames with integer adjacency"; partial frames are not covered. (iv) The transition-DAG definitions (merges, splits, the $(14, 4) \rightleftarrows (14, 5)$ flip-flop) and the 24/16/14 edge counts are supplied [COMPUTED] datums, not formalized.

6 Conclusion

We have proved in Lean the repaired consecutive-firing laws for abstract locks (Theorem 1, [PROVED]), the axiom-free dual-ascent barrier for transition DAGs (Theorem 2, [PROVED]), and the L_s/C3 ascent-depth bounds (Theorem 3, [PROVED]), all with zero sorrys on the bakkie full-library build. The refutations that shaped the repairs — Q-24's $W = \{0, 2\}$ counterexample and Q-26's abstract non-entailment model — are reported honestly and, where possible, promoted into sharpness theorems. What remains open: the complementary halves of the full D7 barrier, the inapplicable conditional D7″, the C1/C2/C4 fragments (all OBJECTION), and the census application of D7′ (Q-33 OBJECTION stands until full-spectrum endpoints or a proved all-duals ascent is available).

7 References

References

[1] P. Erdős, A. Rényi, and V. T. Sós, *On a problem of graph theory*, Studia Sci. Math. Hungar. **1** (1966), 215–235. (The friendship theorem, cited as given among the program's classical references, R298.)

[2] P. Erdős and P. Turán, *On a problem of Sidon in additive number theory*, J. London Math. Soc. **16** (1941), 212–215. (Cited as given among the program's classical Sidon references.)

[3] OpenAI, *github.com/openai/math* catalogue, 719 manuscripts fetched 2026-10-10, 0 "Sidon" hits (cited as given, R273/R279 novelty checks; the σ-spectrum and its transition DAG are program-defined objects).

[4] D. Blair, *Unsolved Math — Round 280: abstract lock theory*, program dossier, 2026-10-10. (Supplied datum: lock-frame definitions, Q-24 OBJECTION, concrete instances, 472-frame census design.)

[5] D. Blair, *Unsolved Math — Round 287: repaired lock laws*, program dossier, 2026-10-11. (Supplied datum: T3(iii′) repair, quantifier-slip mechanism, Q-32 verdict, [COMPUTED]×2 census.)

[6] D. Blair, *Unsolved Math — Round 279: transition DAG census*, program dossier, 2026-10-10. (Supplied datum: $G = (C, E_d \cup E_p \cup \mathcal{L})$ census, 24/16/14 edge counts, merges/splits, flip-flop.)

[7] D. Blair, *Unsolved Math — Round 288: D7′ dual-ascent barrier*, program dossier, 2026-10-11. (Supplied datum: Q-26 model, broken-step location, D7′ proofs, $\mathrm{D7}'/H/H_{\mathrm{sym}}$ audits, Q-33 verdict.)

[8] D. Blair, *Unsolved Math — Round 298: queue hardening*, program dossier, 2026-10-10. (Supplied datum: md5 sync records, L_s/C3 fold, Q-36 verdict.)

[9] D. Blair, *Unsolved Math — Round 310: full-library build gate*, program dossier, 2026-10-11. (Supplied datum: bakkie `lake build` "Build completed successfully (17936 jobs)", exit 0, 0 errors, 0 sorrys; label moves.)